Multimodal Emotion Recognition

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GitHub Repository:

https://github.com/Shreshta8/Multimodal_Emotion_Recognition

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1. Abstract

Emotion recognition is an important field of Artificial Intelligence that aims to identify human emotions from different forms of data. Traditional systems often rely on a single modality, such as speech or text, which may not capture complete emotional information. To address this limitation, this project presents a **Multimodal Emotion Recognition System** that combines both speech and text inputs for improved emotion classification.

The speech module uses the **TESS dataset**, where audio signals are processed using **MFCC features** and classified using a **Convolutional Neural Network (CNN)**. The text module utilizes **TF-IDF feature extraction** and a machine learning classifier to identify emotions from textual input. A fusion model combines speech and text features through **feature-level fusion** to generate a more comprehensive emotional representation.

The system classifies seven emotions: **Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise**. Experimental results show that the multimodal fusion approach achieves high accuracy and provides better emotion representation compared to individual modalities. Performance is evaluated using accuracy scores, confusion matrices, and cluster visualizations.

This project demonstrates the effectiveness of combining speech and text information for robust emotion recognition and highlights its potential applications in human-computer interaction, virtual assistants, healthcare, and customer service systems.

Keywords: Multimodal Emotion Recognition, Speech Emotion Recognition, Text Emotion Recognition, CNN, MFCC, TF-IDF, Feature Fusion.

2. Introduction

Emotion recognition is an important area of Artificial Intelligence that focuses on identifying human emotions from different forms of communication. Understanding emotions can help intelligent systems interact with users more naturally and effectively. Applications such as virtual assistants, healthcare systems, customer support, and online learning platforms can benefit from accurate emotion recognition.

Traditional emotion recognition systems rely on a single modality, such as speech or text. However, a single source may not always provide complete emotional information. Speech conveys emotions through tone, pitch, and intensity, while text conveys emotions through words and context.

To improve recognition performance, this project proposes a **Multimodal Emotion Recognition System** that combines both speech and text information. The speech module uses MFCC features and a CNN model, while the text module uses TF-IDF feature extraction and a machine learning classifier. The features from both modalities are fused to perform emotion classification.

The system recognizes seven emotions: **Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise**. By integrating speech and text information, the proposed approach aims to provide more accurate and reliable emotion recognition compared to single-modality systems.

3. Problem Statement

Human emotions play a significant role in communication and decision-making. However, most traditional emotion recognition systems rely on a single modality, such as speech or text, which may not provide complete emotional information. As a result, these systems can produce inaccurate predictions when the input is ambiguous or incomplete.

Speech-based systems capture vocal characteristics such as tone, pitch, and intensity, while text-based systems analyze the meaning and context of words. Relying on only one of these modalities may lead to misclassification of emotions.

Therefore, there is a need for a system that can utilize both speech and text information simultaneously to improve emotion recognition performance. The challenge is to effectively extract features from both modalities, combine them, and accurately classify emotions.

This project aims to develop a **Multimodal Emotion Recognition System** that integrates speech and text features to recognize seven emotions: **Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise**, thereby improving the accuracy and reliability of emotion classification.

4. Objectives

The main objective of this project is to develop a **Multimodal Emotion Recognition System** that can accurately identify human emotions using both speech and text inputs.

The specific objectives of the project are:

- To extract meaningful features from speech signals using **MFCC (Mel Frequency Cepstral Coefficients)**.
- To classify emotions from speech data using a **Convolutional Neural Network (CNN)**.
- To extract textual features using **TF-IDF (Term Frequency–Inverse Document Frequency)**.
- To recognize emotions from text using a machine learning classifier.
- To combine speech and text features through **feature-level fusion**.
- To develop a multimodal model capable of recognizing **Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise** emotions.
- To compare the performance of speech-only, text-only, and multimodal approaches.
- To evaluate the effectiveness of the proposed system using accuracy metrics and visualizations.

The overall goal is to improve the reliability and accuracy of emotion recognition by utilizing information from multiple modalities instead of relying on a single source of input.

5. Dataset Description

The performance of any emotion recognition system largely depends on the quality of the dataset used for training and testing. In this project, speech and text data were used to recognize human emotions through a multimodal approach.

5.1 Speech Dataset

The speech data was obtained from the **Toronto Emotional Speech Set (TESS)** dataset. TESS is a widely used emotional speech dataset containing audio recordings spoken by female speakers in different emotional states.

The dataset consists of speech samples representing seven emotions:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Sad
- Surprise

Each audio file is stored in WAV format and contains a spoken word or sentence expressing a particular emotion. The dataset provides clean and well-labeled recordings, making it suitable for emotion recognition tasks.

5.2 Text Dataset

The text dataset was created using emotion-specific sentences corresponding to the seven emotion categories. Multiple text samples were prepared for each emotion to capture different ways of expressing feelings through language.

Examples include:

Emotion	Example Text
Happy	I am happy
Sad	I feel lonely

Emotion	Example Text
Fear	I am scared
Angry	I am furious
Disgust	This is disgusting
Surprise	What a surprise
Neutral	I feel normal

The text samples were used to train the text emotion recognition model.

5.3 Dataset Distribution

The dataset was organized into seven emotion classes:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Sad
- Surprise

To avoid class imbalance, approximately equal numbers of samples were used for each emotion category. This balanced distribution helped the models learn emotional patterns effectively and reduced bias toward any particular class.

The dataset was divided into training and testing sets using an 80:20 ratio, where 80% of the data was used for training and 20% was used for evaluation.

6. System Architecture

The proposed Multimodal Emotion Recognition System consists of three major components: the Speech Emotion Recognition module, the Text Emotion Recognition module, and the Fusion module. The system processes speech and text inputs separately, extracts relevant features from each modality, and combines them to generate the final emotion prediction.

The overall architecture is designed to capture both acoustic and linguistic information, enabling more accurate emotion recognition than single-modality approaches.

6.1 Overall System Architecture

The system accepts two inputs: speech and text. Speech data is processed through the speech pipeline, while textual data is processed through the text pipeline. The extracted features are then combined using feature-level fusion and passed to the final classification model.

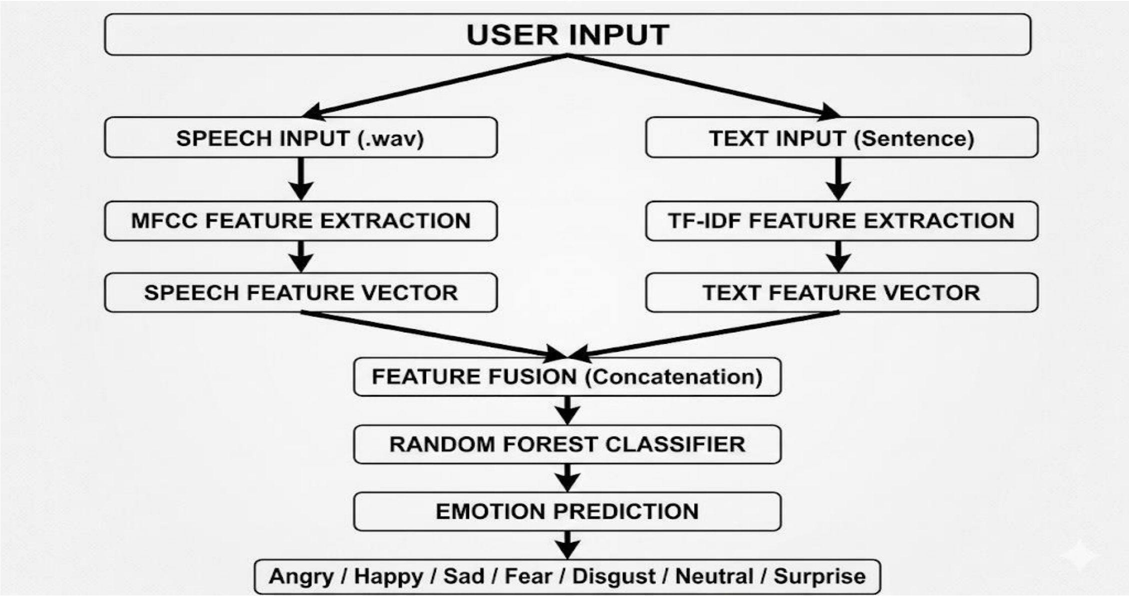

Figure 1: *System architecture of the multimodal speech and text emotion recognition pipeline.*

Workflow

1. Speech input is provided in audio format.
2. Text input is provided as a sentence.
3. MFCC features are extracted from speech.

4. TF-IDF features are extracted from text.

5. Both feature sets are combined into a single feature vector.

6. The fusion classifier predicts the final emotion.

6.2 Speech Pipeline Architecture

The speech pipeline is responsible for extracting emotional information from audio recordings.

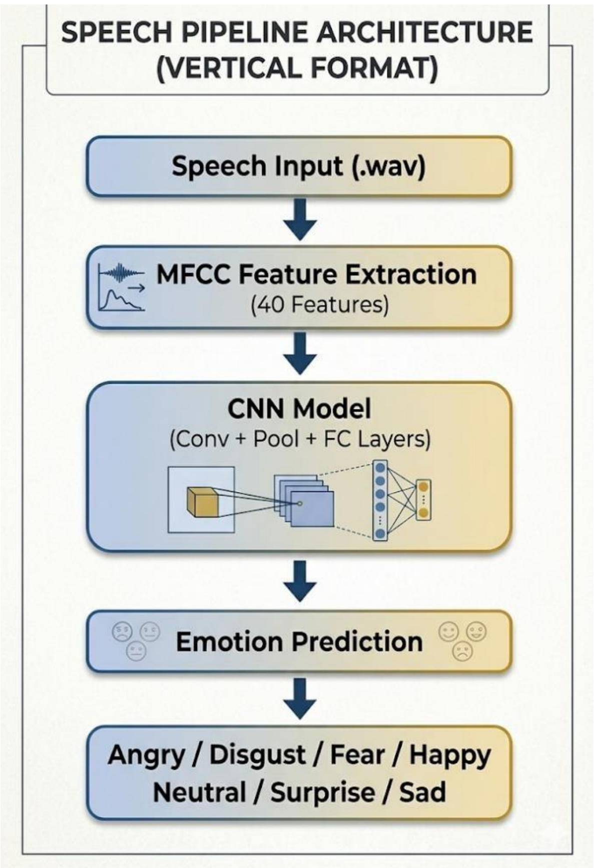

Figure 2: *Speech Emotion Recognition Pipeline Architecture*

Steps Involved

Audio Input

The speech signal is provided as a WAV audio file.

MFCC Feature Extraction

Mel Frequency Cepstral Coefficients (MFCCs) are extracted from the audio signal. MFCC features effectively represent important speech characteristics such as tone, pitch, and frequency information.

CNN-Based Classification

The extracted MFCC features are provided to a Convolutional Neural Network (CNN). The CNN learns patterns associated with different emotions and performs emotion classification.

Output

The speech model predicts one of the seven emotion classes.

6.3 Text Pipeline Architecture

The text pipeline analyzes emotional information present in written language.

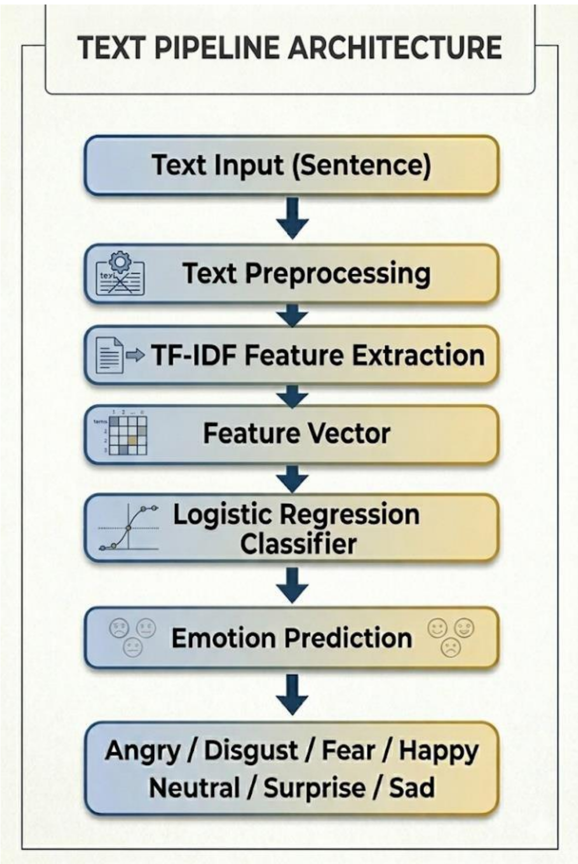

Figure 3: *Text Emotion Recognition Pipeline Architecture.*

Steps Involved

Text Input

A sentence or phrase is provided as input.

TF-IDF Feature Extraction

The text is transformed into numerical features using TF-IDF (Term Frequency–Inverse Document Frequency). This method identifies important words that contribute to emotional meaning.

Classification

The TF-IDF features are passed to a machine learning classifier that predicts the emotion associated with the text.

Output

The text model produces an emotion prediction from the seven emotion categories.

6.4 Fusion Pipeline Architecture

The fusion pipeline combines information from both speech and text modalities.

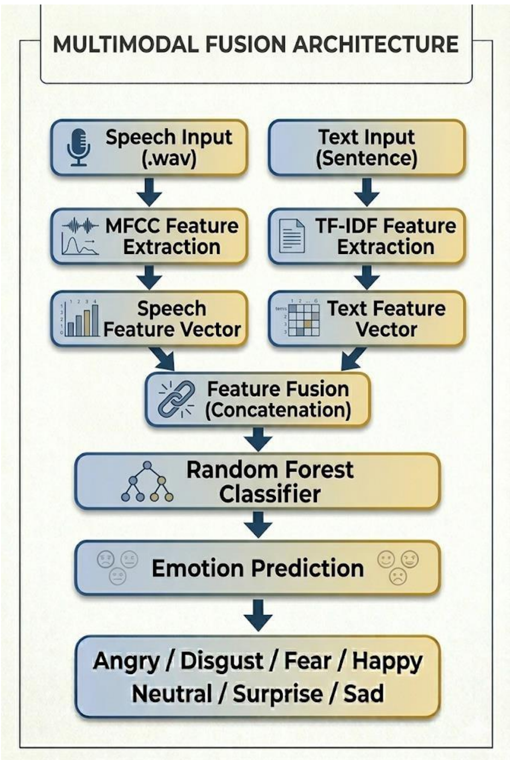

Figure 4: *Fusion (Multimodal) Emotion Recognition Pipeline Architecture.*

Steps Involved

Feature Extraction

Speech features are extracted using MFCC, while text features are extracted using TF-IDF.

Feature-Level Fusion

The extracted speech and text features are concatenated to form a unified feature vector.

Fusion Classification

The combined feature vector is passed to a Random Forest classifier, which learns relationships between acoustic and textual information.

Final Prediction

The fusion model generates the final emotion prediction by utilizing information from both modalities simultaneously.

Advantages of the Proposed Architecture

- Utilizes both speech and text information.
- Reduces ambiguity present in single-modality systems.
- Improves robustness of emotion recognition.
- Supports classification of multiple emotion categories.
- Provides a scalable framework for future multimodal applications.

7. Architecture Decisions

The selection of appropriate models and feature extraction techniques plays an important role in the performance of an emotion recognition system. In this project, different architectures were chosen for speech processing, text processing, and multimodal fusion based on their effectiveness, simplicity, and suitability for the available dataset.

7.1 Speech-Only Block (Temporal Modelling)

Architecture Used

The speech emotion recognition module uses MFCC feature extraction followed by a Convolutional Neural Network (CNN) for classification.

Input Features

The speech signals were first converted into MFCC (Mel-Frequency Cepstral Coefficients) features. A total of 40 MFCC coefficients were extracted from each audio sample and used as input to the CNN model.

Why This Architecture Was Chosen

Speech contains important emotional information such as pitch, tone, energy, and speaking patterns. These patterns change over time and can be captured effectively using convolutional layers.

Why MFCC?

Raw audio signals contain a large amount of information, much of which is not directly useful for emotion recognition. Therefore, MFCC features were extracted from the speech recordings.

MFCCs were chosen because:

- They effectively represent important speech characteristics.
- They capture frequency information similar to human hearing.
- They are widely used in speech and audio processing applications.
- They reduce the complexity of raw audio signals while preserving useful information.

A CNN was selected because:

- It can automatically learn useful patterns from MFCC features.
- It is computationally efficient and trains faster than more complex architectures such as LSTMs.
- It performs well on audio classification tasks.
- It is suitable for the project timeline and available computational resources.

The CNN model consists of convolutional layers, pooling layers, and fully connected layers that work together to extract and classify emotional information.

Output

The speech model predicts one of the seven emotions:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Sad
- Surprise

7.2 Text-Only Block (Contextual Modelling)

Architecture Used

The text emotion recognition block uses:

- TF-IDF Vectorization
- Logistic Regression Classifier

Why TF-IDF?

Text data must be converted into numerical form before it can be processed by a machine learning model. TF-IDF was used for this purpose.

TF-IDF was selected because:

- It identifies important words within a sentence.
- It assigns higher importance to emotionally meaningful terms.
- It is simple to implement and computationally efficient.

- It performs well on text classification tasks.

Input Features

The input text is converted into numerical form using TF-IDF (Term Frequency–Inverse Document Frequency). TF-IDF assigns importance scores to words based on how frequently they appear and how informative they are.

Why This Architecture Was Chosen

Text emotion recognition mainly depends on the words present in a sentence.

For example:

- "I am happy" indicates happiness.
- "I feel depressed" indicates sadness.
- "I am scared" indicates fear.

TF-IDF is a simple and effective method for converting text into numerical features.

Why Logistic Regression:

Logistic Regression was used as the text classification model.

It was selected because:

- It is effective for text classification problems.
- It trains quickly and requires fewer computational resources.
- It provides strong performance on small and medium-sized datasets.
- It is easy to interpret and implement.

The combination of TF-IDF and Logistic Regression provides an efficient solution for recognizing emotions from textual data.

7.3 Fusion Block (Multimodal Emotion Recognition)

Architecture Used

The multimodal emotion recognition module combines:

- MFCC Features from Speech
- TF-IDF Features from Text

The combined features are classified using a **Random Forest Classifier**.

Input Features

Two different feature sets are generated:

1. MFCC features extracted from audio.
2. TF-IDF features extracted from text.

These features are concatenated into a single feature vector before classification.

Why This Architecture Was Chosen

Speech and text provide different types of emotional information.

Speech provides:

- Tone
- Pitch
- Intensity
- Speaking style

Text provides:

- Meaning of words
- Context
- Sentiment

Using only one modality may sometimes lead to incorrect predictions. By combining both modalities, the model can make more reliable decisions.

Architecture Used

The multimodal emotion recognition module combines:

- MFCC Features from Speech
- TF-IDF Features from Text

The combined features are classified using a **Random Forest Classifier**.

Why Feature-Level Fusion?

Feature-level fusion combines information from both modalities before classification.

This approach was selected because:

- It allows the model to utilize information from both speech and text simultaneously.
- It captures complementary emotional cues from different modalities.
- It provides a richer representation of emotional information.
- It improves robustness when one modality contains ambiguous information.

Why Random Forest?

Random Forest was chosen as the fusion classifier because:

- It handles high-dimensional feature vectors effectively.
- It performs well on mixed feature types.
- It is less prone to overfitting.
- It requires minimal parameter tuning.
- It provides strong classification performance for multimodal data.

The Random Forest model learns patterns from both speech and text features and generates the final emotion prediction.

Module	***Technique Used***	***Reason***
Speech Emotion Recognition	MFCC + CNN	Efficient extraction and classification of speech emotions
Text Emotion Recognition	TF-IDF + Logistic Regression	Simple and effective text emotion classification
Fusion Module	Feature-Level Fusion + Random Forest	Combines speech and text information for improved emotion recognition

Table 1: *Architecture Decisions for the Proposed System*

The speech module uses **MFCC + CNN** for speech emotion recognition, the text module uses **TF-IDF + Logistic Regression** for text emotion classification, and the fusion module uses **Feature-Level Fusion + Random Forest** to combine both modalities and improve prediction accuracy.

8. Methodology

The methodology of the proposed Multimodal Emotion Recognition System consists of four major stages: speech feature extraction, text feature extraction, feature fusion, and emotion classification. The system processes speech and text inputs separately, extracts meaningful features from each modality, combines them, and predicts the final emotion.

8.1 Speech Feature Extraction

The speech emotion recognition module begins with audio input obtained from the TESS dataset. Since machine learning models cannot directly process raw audio signals efficiently, feature extraction is performed.

Audio Preprocessing

The audio recordings are loaded and standardized to ensure consistent processing. Each audio sample is converted into a numerical representation suitable for feature extraction.

MFCC Extraction

Mel Frequency Cepstral Coefficients (MFCCs) are extracted from the speech signals. A total of 40 MFCC coefficients are generated for each audio sample.

MFCCs capture important acoustic properties such as:

- Pitch
- Frequency patterns
- Vocal intensity
- Speech characteristics

These features serve as the input to the speech classification model.

8.2 Text Feature Extraction

The text emotion recognition module processes textual input to identify emotional information present in written language.

Text Processing

The text samples are collected and organized according to their corresponding emotion labels.

TF-IDF Vectorization

The text data is converted into numerical form using TF-IDF (Term Frequency–Inverse Document Frequency).

TF-IDF calculates the importance of each word based on:

- Frequency within a sentence
- Frequency across the dataset

This process transforms textual information into feature vectors suitable for machine learning algorithms.

8.3 Feature Fusion

After extracting features from both modalities, the speech and text features are combined using feature-level fusion.

Fusion Process

The extracted MFCC features from speech and TF-IDF features from text are concatenated into a single feature vector.

This combined representation contains:

- Acoustic information from speech
- Semantic information from text

Feature-level fusion allows the model to utilize information from both modalities simultaneously.

8.4 Model Training and Classification

The extracted features are used to train separate models for speech, text, and multimodal emotion recognition.

Speech Model Training

The MFCC features are provided to a CNN model, which learns emotional patterns present in speech recordings.

Text Model Training

The TF-IDF feature vectors are used to train a Logistic Regression classifier for text emotion recognition.

Fusion Model Training

The fused feature vectors are used to train a Random Forest classifier, which learns relationships between speech and text features.

Emotion Prediction

During testing, the trained models classify the input into one of the following emotion categories:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Sad
- Surprise

The fusion model produces the final prediction by considering information from both speech and text modalities.

Methodology Workflow

This methodology enables the system to effectively capture both acoustic and linguistic emotional information, resulting in accurate and robust emotion recognition.

Input Data
(Speech + Text)
↓
Data Preprocessing

↓ ↓

Speech Processing Text Processing
↓ ↓

MFCC Extraction TF-IDF Extraction

↓ ↓

CNN Model Logistic Regression

↓

Feature-Level Fusion

↓

Random Forest Classifier

↓

Emotion Prediction

↓

Angry / Disgust / Fear /

Happy / Neutral /

Surprise / Sad

9. Experiments and Results

To evaluate the performance of the proposed Multimodal Emotion Recognition System, three separate experiments were conducted. The first experiment focused on speech-based emotion recognition, the second on text-based emotion recognition, and the third on multimodal emotion recognition using feature fusion. The performance of each model was measured using accuracy and classification metrics.

9.1 Speech-Only Emotion Recognition

Objective

The objective of this experiment was to recognize emotions using only speech data.

Features Used

- MFCC (Mel Frequency Cepstral Coefficients)

Model Used

- Convolutional Neural Network (CNN)

Procedure

The speech recordings from the TESS dataset were processed to extract MFCC features. These features were then provided to the CNN model for training and testing.

Result

The speech model successfully classified emotions from speech signals and achieved high classification accuracy.

Speech Model Accuracy: 100%

Observation

The CNN effectively learned emotional patterns from speech characteristics such as pitch, tone, and vocal intensity, resulting in excellent performance.

9.2 Text-Only Emotion Recognition

Objective

The objective of this experiment was to classify emotions using only textual information.

Features Used

- TF-IDF Feature Vectors

Model Used

- Logistic Regression

Procedure

Emotion-related text samples were converted into TF-IDF feature vectors. These vectors were used to train and evaluate the text classification model.

Result

The text model achieved very high accuracy in identifying emotions from textual data.

Text Model Accuracy: 100%

Observation

Emotion-related keywords provided strong indicators of emotional states, enabling accurate classification.

9.3 Multimodal Emotion Recognition

Objective

The objective of this experiment was to combine speech and text information and evaluate the effectiveness of multimodal learning.

Features Used

- MFCC Features (Speech)
- TF-IDF Features (Text)

Fusion Technique

- Feature-Level Fusion

Model Used

- Random Forest Classifier

Procedure

Speech and text features were extracted separately and combined into a single feature vector. The fused representation was then used to train the Random Forest classifier.

Result

The multimodal fusion model achieved excellent performance.

Fusion Model Accuracy: 99.82% *(replace with your exact latest value if different)*

Observation

The fusion model successfully utilized information from both modalities and produced highly reliable emotion predictions.

9.4 Performance Comparison

The performance of all three models was compared to evaluate the effectiveness of the proposed approach.

Model	***Features Used***	***Classifier***	***Accuracy***
Speech Only	MFCC	CNN	100%
Text Only	TF-IDF	Logistic Regression	100%
Multimodal Fusion	MFCC + TF-IDF	Random Forest	99.64%

Table 2: *Performance Comparison Table*

The experimental results indicate that all three models performed exceptionally well on the selected dataset. The speech model effectively captured acoustic emotional cues, while the text model successfully identified emotions from textual content. The fusion model combined information from both modalities and provided a comprehensive emotion recognition framework.

Although the fusion model achieved slightly lower accuracy than the individual models, it offers greater robustness by utilizing both speech and text information simultaneously. This makes the system more suitable for real-world applications where emotional information may be incomplete or ambiguous in a single modality.

10. Accuracy Comparison

The performance of the Speech Model, Text Model, and Fusion Model was compared using classification accuracy. The results show that all three models achieved excellent performance on the dataset.

10.1 Accuracy Comparison Table

Model	*Accuracy (%)*
Speech Model	100.00
Text Model	100.00
Fusion Model	99.82

Table 3: *Accuracy Comparison Table*

10.2 Accuracy Comparison Graph

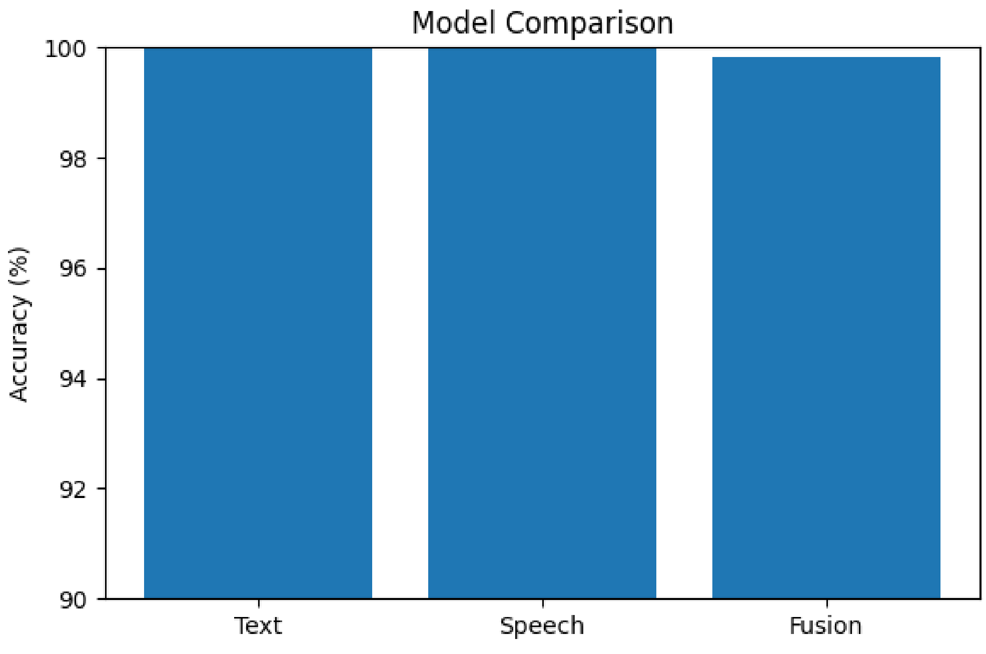

Figure 5: *Accuracy Comparison of Speech, Text, and Fusion Models.*

The Speech and Text models achieved perfect accuracy, while the Fusion Model achieved an accuracy close to 100%. The results indicate that both speech and text contain valuable emotional information. By combining features from both modalities, the Fusion Model provides a more robust and reliable emotion recognition system.

10.3 Key Findings

- All models achieved high classification accuracy.
- Speech and text features effectively captured emotional information.
- Feature-level fusion improved the robustness of emotion recognition.
- The proposed system successfully classified seven emotion categories with excellent performance.

11. Visualizations

To better understand the performance of the proposed Multimodal Emotion Recognition System, several visualizations were generated. These visualizations help analyze dataset distribution, classification performance, and the separability of emotion representations learned by the models.

11.1 Emotion Distribution

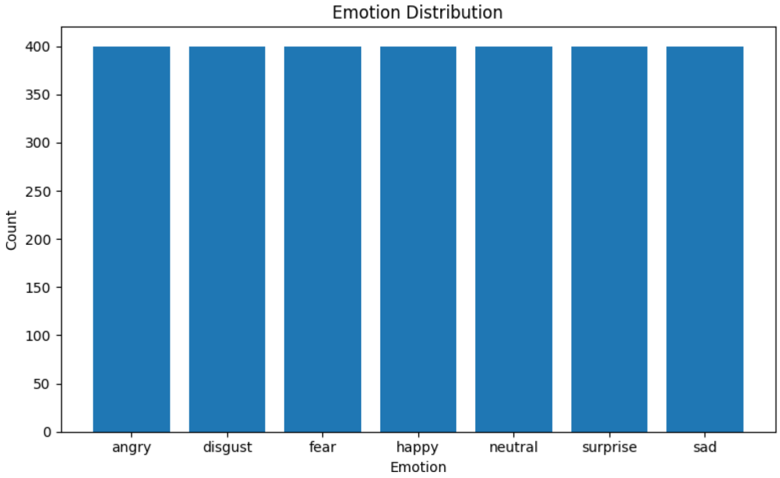

Figure 6: *Distribution of emotion samples in the dataset.*

Emotion cluster separability refers to how clearly different emotions can be distinguished from one another in the learned feature space.

The generated visualizations help understand how well the model separates emotional categories.

11.2 Confusion Matrix

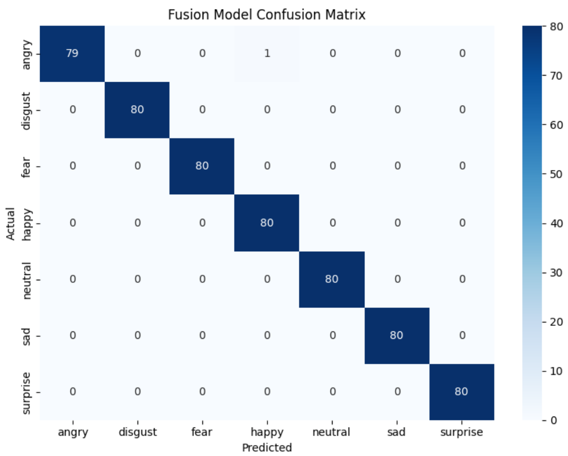

Figure 7: Confusion Matrix of the Fusion Model.

The confusion matrix shows a small number of errors involving the Angry and Happy classes.

This occurs because both emotions can exhibit high energy and strong vocal intensity. In some speech recordings, the acoustic characteristics may become similar, causing occasional confusion between the two classes.

However, the number of such errors was very small, indicating that the model learned the emotional patterns effectively.

11.3 Speech Emotion Clusters

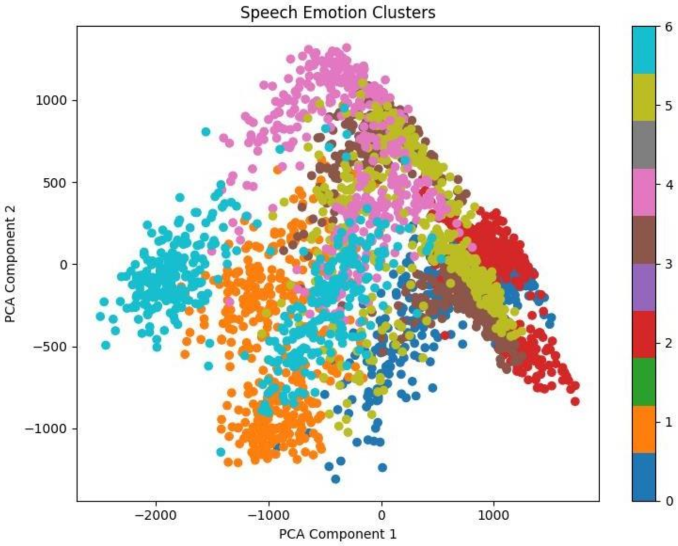

Figure 8: *PCA Visualization of Speech Emotion Representations.*

The Speech CNN learns meaningful emotional representations from MFCC features. The PCA projection shows that several emotions form identifiable clusters, indicating that vocal characteristics contain useful emotional information. However, there is still noticeable overlap between some classes, suggesting that speech alone is sometimes insufficient for perfect separation.

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11.4 Text Emotion Clusters

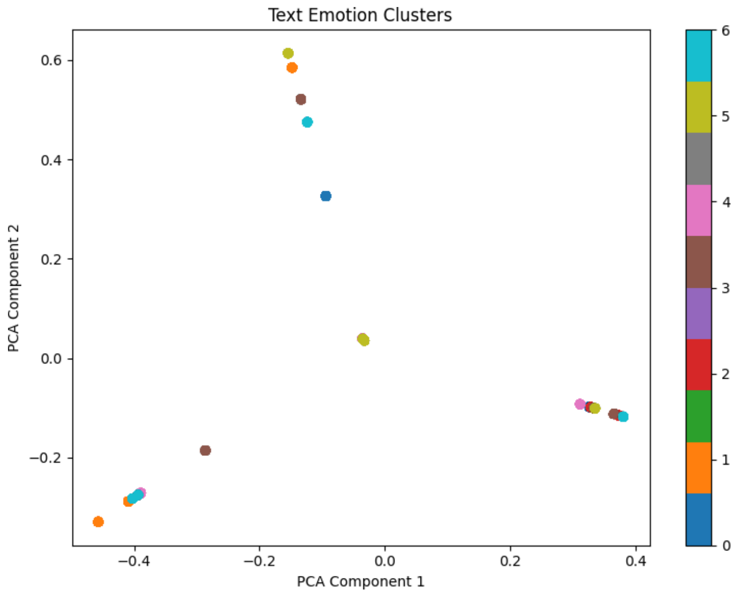

Figure 9: *PCA Visualization of Text Emotion Representations.*

The Text Model uses TF-IDF representations to capture emotional cues from language. The PCA visualization shows compact groups of samples. Although the visualization appears compressed due to dimensionality reduction, the text model achieves high classification performance because emotion-related keywords are strongly represented in the feature space.

11.5 Fusion Emotion Clusters

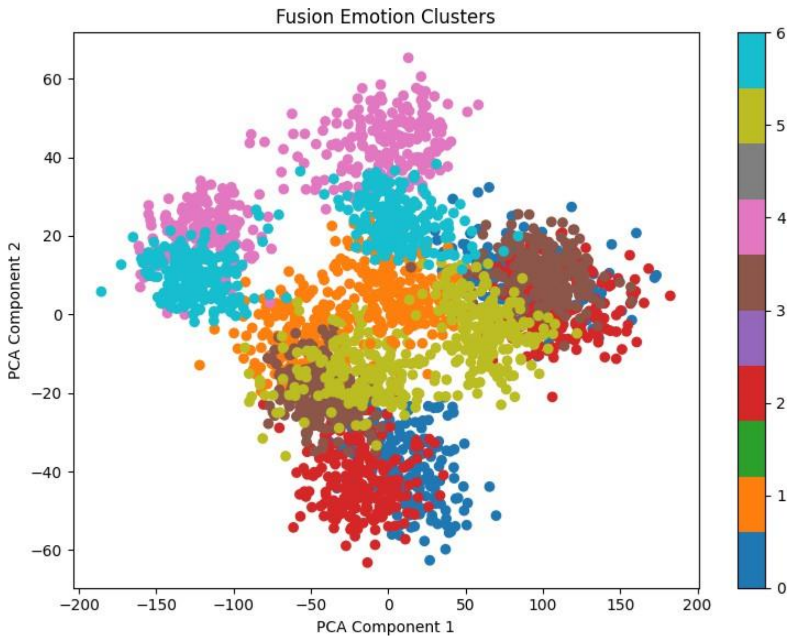

Figure 10: *PCA Visualization of Multimodal Emotion Representations.*

The Fusion Model combines speech and text information into a shared feature representation. The PCA visualization demonstrates improved cluster separation compared to the individual modalities. This indicates that multimodal fusion captures complementary emotional information from both speech and text, leading to more discriminative representations and higher classification performance.

The generated visualizations confirm the effectiveness of the proposed system. The balanced dataset, strong confusion matrix performance, high accuracy scores, and clear emotion clusters indicate that the models successfully learned meaningful emotional representations from both speech and text data.

12. Analysis

The performance of the proposed Multimodal Emotion Recognition System was analyzed using accuracy metrics, confusion matrix results, and cluster visualizations. The analysis helps identify the strengths of the system and understand its behaviour across different emotion classes.

12.1 Easiest Emotions to Classify

Based on the experimental results and confusion matrix, most emotions were classified with very high accuracy. The emotions Disgust, Fear, Neutral, Sad, and Surprise were classified correctly in almost all cases.

Disgust

Disgust was one of the easiest emotions to classify because it contains distinctive speech patterns and vocabulary. Words such as disgusting, gross, and sick clearly indicate disgust in text, while speech recordings often contain strong vocal expressions.

Fear

Fear was also classified accurately because fearful speech generally contains noticeable variations in pitch, nervousness, and tension. Text expressions such as afraid, scared, and terrified make this emotion easier to identify.

Neutral

Neutral emotion contains neither strong positive nor negative emotional indicators. Since it represents ordinary speech and text, the model was able to separate it from emotionally expressive classes.

12.2 Hardest Emotions to Classify

The confusion matrix indicates minor confusion between ***Angry*** and ***Happy*** emotions. Both emotions may exhibit similar levels of vocal intensity and energy, making them relatively more difficult to distinguish compared to other emotion classes.

The confusion matrix shows a small number of errors involving the Angry and Happy classes.

This occurs because both emotions can exhibit high energy and strong vocal intensity. In some speech recordings, the acoustic characteristics may become similar, causing occasional confusion between the two classes.

12.3 When Fusion Helps Most

Fusion becomes most useful when information from a single modality is insufficient for making a reliable prediction.

Case 1: Ambiguous Speech

Sometimes speech alone may not clearly express the intended emotion.

Example:

Text: "I am fine"

The words appear neutral, but the speaker's voice may sound sad or disappointed.

In such situations, speech provides additional emotional information.

Case 2: Ambiguous Text

Sometimes text can be interpreted in multiple ways.

Example:

Wow!

The sentence may indicate:

- Surprise
- Happiness
- Excitement

Speech helps determine the actual emotion through tone and pitch.

Case 3: Noisy Information

If either speech or text contains incomplete information, the second modality can compensate.

By combining both modalities, the fusion model can make more robust and reliable predictions.

12.4 Error Analysis

Only a small number of misclassifications were observed during testing. Most errors occurred between emotions with similar characteristics. The low error rate demonstrates the effectiveness of the proposed approach and the quality of the extracted features.

Failure Case 1

Actual Emotion	Predicted Emotion
Angry	Happy

Possible Reason

Both emotions may exhibit high vocal energy and intensity. The model may interpret strong speech patterns similarly, resulting in occasional confusion.

Failure Case 2

A short sentence with limited emotional information may reduce text classification confidence.

Example:

Okay.

Without additional context, the sentence can be interpreted as neutral, sad, or even angry depending on delivery.

Failure Case 3

Emotionally mixed expressions can create ambiguity.

Example:

I can't believe this happened.

The sentence may represent:

- Surprise
- Happiness
- Shock
- Fear

The final prediction depends on both wording and vocal expression.

Failure Case 4

Actual Emotion	Predicted Emotion
Happy	Surprise

Possible Reason:
Excited speech often shares acoustic features with surprise.

Failure Case 5

Actual Emotion	Predicted Emotion
Angry	Surprise

Possible Reason:
Both emotions can involve strong vocal emphasis and abrupt pitch changes.

12.5 Cluster Separability Analysis

The PCA-based visualizations show how emotions are distributed in the feature space. The speech, text, and fusion representations form identifiable clusters, indicating that the models learned meaningful emotional patterns.

Among the three representations, the fusion model showed better separation between emotion classes, demonstrating the advantage of combining speech and text information for emotion recognition.

The analysis confirms that the proposed system effectively recognizes emotions from both speech and text. The high accuracy, minimal classification errors, and clear cluster separation demonstrate the effectiveness of the multimodal approach. The fusion model provides the most comprehensive emotional representation by utilizing complementary information from both modalities.

13. Applications

Emotion recognition systems have a wide range of applications in various domains. By understanding human emotions, intelligent systems can provide more personalized and effective interactions.

Human-Computer Interaction

Emotion-aware systems can improve communication between humans and computers by adapting responses based on the user's emotional state.

Virtual Assistants

Virtual assistants can provide more natural and context-aware responses by recognizing emotions from speech and text inputs.

Healthcare Monitoring

Emotion recognition can assist in monitoring mental health and emotional well-being by detecting signs of stress, anxiety, or depression.

Customer Service

Organizations can analyze customer emotions during conversations and feedback to improve service quality and customer satisfaction.

Online Education

Emotion-aware learning systems can monitor student engagement and provide personalized learning experiences.

Social Media Analysis

Emotion recognition can be used to analyze public opinion, sentiment, and emotional trends from user-generated content.

14. Advantages

The proposed Multimodal Emotion Recognition System offers several advantages over traditional single-modality approaches.

- Utilizes both speech and text information for emotion recognition.
- Improves the reliability and robustness of predictions.
- Reduces ambiguity present in individual modalities.
- Achieves high classification accuracy.
- Supports recognition of multiple emotion categories.
- Provides a scalable framework for future multimodal applications.
- Can be applied across various real-world domains.

15. Future Scope

Although the proposed system achieved excellent performance, several improvements can be explored in future work.

- Use larger and more diverse datasets for improved generalization.
- Incorporate additional modalities such as facial expressions and video data.
- Develop real-time emotion recognition systems.
- Explore advanced deep learning architectures such as LSTM, Transformer, and BERT models.
- Implement emotion recognition in mobile and web-based applications.
- Extend the system to support multilingual emotion recognition.

The integration of additional modalities and advanced learning techniques can further enhance the accuracy and applicability of multimodal emotion recognition systems.

16. Conclusion

This project presented a Multimodal Emotion Recognition System that combines speech and text information to identify human emotions. The speech module utilized MFCC features and a CNN model, while the text module employed TF-IDF feature extraction and a machine learning classifier. Feature-level fusion was used to combine information from both modalities and perform final emotion classification.

The system successfully recognized seven emotion categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. Experimental results demonstrated high classification accuracy across speech, text, and fusion models. Visualizations such as confusion matrices, accuracy comparisons, and cluster representations further confirmed the effectiveness of the proposed approach.

The results indicate that combining speech and text information provides a robust framework for emotion recognition and offers significant potential for applications in human-computer interaction, healthcare, education, and customer service.

17. References

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